

VIDIT SHAH

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EDUCATION

Birla Institute of Technology and Science Pilani, Pilani Campus, Pilani, Rajasthan **2023 - Present**
Bachelor of Engineering in Mechanical Engineering. CGPA: 8.79
Prakash College of Commerce and Science, Mumbai, Maharashtra **2021 - 2023**
HSC (PCM), French

INTERNSHIPS/EXPERIENCE

Undergraduate Researcher, BITS Pilani, Pilani, Rajasthan **Jan 2025 - Present**
Working as a student researcher under the guidance of Dr. Radha Raman Mishra (Assistant Professor of Mechanical Engineering).

- Conceptualized and constructed a Proof-of-Concept for **Rapid Volumetric Polymerization of Optically Opaque Resin using Microwave Energy**. The process describes volumetric curing of resin, using Microwaves to cure a rotating vat of resin.

Mechanical Engineering Lead, CRISS Robotics, Pilani, Rajasthan **Feb 2025 - Present**
CRISS Robotics is BITS Pilani's premier student technical team, working on development of a Mars Rover Prototype.

- As the Mechanical Engineering Lead, I **led a team of over 25 mechanical engineering students** for development of a Mars rover prototype. We start off with ideation, then we go to CAD, with several iterations, along with integration across the various sub-assemblies, as well as FEA analysis of the designs.

Summer Intern, L&T Heavy Engineering, Larsen & Toubro Group, Surat, Gujarat **May 2025 - July 2025**
L&T Heavy Engineering is a leader in heavy manufacturing, supplying engineered to order equipment for various heavy industries.

- Designed a 6-Axis Robotic Arm from scratch**, with the intended purpose of automating **GTAW Buttering**. The arm had a maximum reach of over 2 meters, with a nominal payload of 10 kilograms. The Process involved end-to-end component selection, vendor outreach, sourcing of gearboxes and motors, CAD Design in SolidWorks, and Finite Element Analysis.

Summer Intern, Ernst & Young (EY), Mumbai, Maharashtra **June 2024 - Aug 2024**
EY is a global professional services organization that provides assurance, consulting, tax, strategy, and transactions services.

- Developed a **multivariate regression model in Python** with libraries such as Pandas & Numpy. The model was able to predict sales, costs and production figures using historical company data.

VOLUNTEERING/SOCIAL WORK

Executive Committee Member, National Service Scheme, BITS Pilani **Aug 2023 - Present**
NSS is a social service initiative by the Government of India, to inculcate social responsibility in the Indian youth.

- Led Umang Collection Drive, **engaging 4000+ students to raise over a total of ₹30L (35,000\$) over 3 years**, funding scholarships and educational support for underprivileged children. The three-day donation drive consisted of unique and marketing challenges, requiring problem solving and people skills, resulting in over 500 scholarships over a 3-year period

RELEVANT COURSEWORK

- Completed Courses:** Mechanics of Solids, Mechanisms & Machines, Advanced Mechanics of Solids, Design of Machine Elements, Additive Manufacturing, Digital Image Processing
- Ongoing/Upcoming Courses (To be completed by May 2026):** Robotics, Control Systems, Vibrations & Control, Computer Aided Design

PROJECTS

5-Axis Robotic Arm

- Developing a 5-axis Robotic Arm. Work includes **complete mechanical and electrical design**, including custom cycloidal gearboxes & absolute encoding as well as a differential wrist.

Soft Robotic End Effector

- Working on a pneumatically actuated **Soft Robotic End Effector**, capable of interacting with soft, fragile & irregular objects.

Continuum Robot

- Leading Development of a **tendon-driven Continuum Robot** for navigating complex obstructions, with theoretically infinite DOFs, inspired by biological manipulators. Used compliant FDM printed, cable-driven segments for actuation.

TECHNICAL SKILLS

CAD (Computer Aided Design): SolidWorks, Fusion 360, Siemens Solid Edge

CAE (Simulation & Analysis): MATLAB, Ansys, Multibody Simscape

LANGUAGES

English: Native/Bilingual

Hindi: Native/Bilingual

Gujarati: Native/Bilingual

French: Elementary Fluency